BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

Consolidated Edison Company of New York, Inc.

Cases 25-E-0072 and 25-G-0073

May 2025

Prepared Corrected Testimony of:
Staff Sales Forecasting Panel

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Q. Would the members of the Staff Sales Forecasting Panel please state your names, employer, and business address?

A. Our names are Richard Schuler, Arslan Gohir, James Creaser, Nathaniel Moneymaker, and Anna Carroll. We are employed by the New York State Department of Public Service (Department). Our business address is Three Empire State Plaza, Albany, New York 12223.

Q. Dr. Schuler, what is your position in the Department?

A. I am employed as a Chief of Regulatory Economics, in the Office of Regulatory Economics.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Arts Degree in Economics from Cornell University in 1985, a Certificate in Regulatory Economics from the State University of New York at Albany in 1989, a Master of Arts in Economics from the State University of New York at Albany in 1991, and a Ph.D. in Managerial Economics from the Rensselaer Polytechnic Institute in 1998. From

1985 to the present I have been employed by the Department.

Q. Have you previously testified before the Public Service Commission, referred to as the Commission?

A. Yes. I testified in Cases 24-E-0461 and 24-G-0462, Case 23-G-0627, Cases 23-E-0418 and 23-G-0419, Case 15-M-0742, Cases 15-M-0127/12-M-0476/98-M-1343, Cases 12-E-0201 and 12-G-0202, Case 11-E-0408, Case 09-M-0527, Case 09-W-0824, Cases 09-S-0029/09-S-0794, Case 96-C-0603, Case 94-C-0017, Case 92-C-0665, Case 92-C-0272, Case 91-C-0123, Case 28425, Case 90-C-0191, Case 89-C-008, Case 89-C-022 and Case 28978. I also submitted affidavits in Cases 13-M-0026 and 10-M-0451.

Q. Mr. Gohir, what is your position in the department?

A. I am employed as an Associate Economist in the Office of Regulatory Economics.

Q. Please describe your educational background.

A. I received a Bachelor of Arts, Specialized Honors in Financial and Business Economics from York University in 2016 and the Master of Arts

in Economics in 2019 from York University. Prior to joining the Department, I worked at Xcel Energy Inc. from 2020 to 2023 in the Energy Sales and Demand Forecasting Group and the Pricing and Planning team. Previously, I worked at the Independent Electricity System Operator of Ontario from 2016 to 2017 as a Reporting Specialist in the Conservation Intelligence and Analytics team. In 2015, I worked at the Hydro One Network's Economic and Load Forecasting group. Finally, I completed a one-year internship at the Toronto Hydro Inc. where I worked in the Secondary Distribution Services group.

Q. Please briefly describe your current responsibilities at the Department.

A. My main responsibilities as an Associate Economist include analyzing economic issues related to public utility regulation such as econometric sales forecasting via the specification and estimation of econometric models of customer demand. Testifying as an expert witness in administrative hearings in support of the reasonableness of those sales

models and forecasts.

Q. Have you previously testified before the Commission?

A. Yes. I have testified before the Commission in the rate proceedings regarding sales and customer forecasting. Most recently I have testified in Cases 24-E-0461 and 24-G-0462, Cases 24-E-0060 & 24-G-0061, 24-E-0322 & 24-G-0323. Additionally, during my previous employment at Xcel Energy Inc. I prepared electric sales, customer count, and peak demand forecasts for Xcel's 2022 New Mexico rate case; weather normalized sales impact work papers for retail electric sales, and peak demand forecasts for Xcel's Texas 2023 and 2021 rate cases. I have also assisted in Jurisdictional Cost of Service Studies (JCOSS), Class Cost of Service Studies (CCOSS), and Rate Design studies for Xcel's 2022 New Mexico and 2023 Texas rate cases. Additionally, I have filed testimony for Xcel's subsidiary Southwestern Public Service, (or SPS), Company's Energy Efficiency and Cost Recovery Factor (EECRF) filing before the Texas Commission. I have also assisted with the

electric sales forecast, peak demand, and load analysis studies in the 2021 SPS New Mexico Integrated Resource Plan.

Q. Mr. Creaser, what is your position in the department?

A. I am employed as a Senior Economist in the Office of Regulatory Economics

Q. Please describe your educational background.

A. I received a Bachelor of Science in Business Economics and Communications from the State University of New York at Oneonta in 2020. I received a Master of Science in Economics from Purdue University in 2025. From May 2020 through March 2022, I worked at Homestead Funding Corporation as a loan officer. I have been employed by the Department since 2022.

Q. Have you previously testified before the Commission?

A. Yes. I have testified before the Commission in many rate proceedings. Most recently I testified in Cases 22-E-0317 and 22-G-0318, regarding New York State Electric and Gas; Cases 22-E-0319 and 22-G-0320, regarding Rochester Gas and Electric; Case 22-S-0659, regarding

Consolidated Edison Company of New York, Inc.; Cases 23-G-0225 and 23-G-0226, regarding Keyspan Energy Delivery New York and Keyspan Energy Delivery Long Island, respectively; Cases 23-E-0418 and 23-G-0419, regarding Central Hudson; Case 23-G-0627 regarding National Fuel Gas Distribution Corporation; Cases 24-E-0060 and 24-G-0061, regarding Orange & Rockland Utilities, Inc.; Cases 24-E-0322 and 24-G-0323, regarding Niagara Mohawk Power Company d/b/a National Grid; and Case 24-G-0447, regarding Corning Natural Gas Corporation.

Q. Mr. Moneymaker, what is your position in the department?

A. I am employed as a Senior Economist in the Office of Regulatory Economics

Q. Please describe your educational background

A. I received a Bachelor of Science in Environmental and Resource Economics, as well as a Master of Science in Environmental Economics from the University of Delaware in May 2022 and August 2024 respectively. I have been employed by the Department since 2025. Prior to my employment by the Department, I worked through

grant funding under an assistantship from May 2022 to December 2024. This work involved econometric estimation of per trip values for recreational saltwater fishing in Delaware.

Q. Have you previously testified before the Commission?

A. I have not previously testified before the Commission.

Q. Ms. Carroll, what is your position in the department?

A. I am employed as an Economist Trainee 1 in the Office of Regulatory Economics.

Q. Please describe your educational background.

A. I earned my Bachelor of Science in Business Administration concentrating in Economics from Appalachian State University in May of 2024. I have been employed by the Department since 2024.

Q. Have you previously testified before the Commission?

A. Yes, I testified in Cases 24-E-0461 and 24-G-0462, regarding Central Hudson Gas and Electric Corporation.

Q. What is the purpose of the Panel’s testimony in these proceedings?

A. Our testimony will provide our recommendations regarding the electric and gas customer count and delivery volume forecasts for Con Edison, in response to testimony filed by the Company's Electric and Gas Forecasting Panels, referred to as the EFP and GFP, respectively. We will also address the additional forecasting related Full Time Equivalent (FTE) employees and additional forecast related Operations and Maintenance (O&M) expenses requested by the EFP and GFP.

Q. In your testimony, will you refer to, or otherwise rely upon, any information produced during the discovery phase of these proceedings?

A. Yes. We will refer to, and have relied upon, several responses to Staff Information Requests (IRs). These responses are included in Exhibit___(SSFP-1) and are referenced by the designation given to the IR by Staff.

Q. Is the Panel sponsoring any other exhibits?

A. Yes. We are also sponsoring seven additional exhibits.

Q. Please briefly describe these exhibits.

A. Exhibit___(SSFP-2) provides a summary of the Panel's forecast for electric delivery volume

and customer counts, along with a comparison with the Company's forecast. Corrected Exhibit___(SSFP-3) is a summary of the Panel's gas volume and customer count forecasts, along with a comparison with the Company's gas forecast. Exhibit___(SSFP-4) and Exhibit___(SSFP-5) show the output of the Panel's electric and gas forecasting models, respectively. Exhibit___(SSFP-6) presents our analyses of post-model adjustments for the electric volume forecast. Exhibit___(SSFP-7) provides our analysis of the gas energy efficiency post-model adjustment. Finally, Exhibit___(SSFP-8) presents the results of our aggregate, top-down forecast modelling.

FORECAST SUMMARY

Q. Please describe the Company's electric delivery volume forecast.

A. On page 3 of the EFP's updated testimony, the Company forecasts an electric delivery volume of 53,319 gigawatt hours (GWh) for the Rate Year, the 12 months ending December 31, 2026.

Q Do you agree with the Company's electric forecast?

A. No, we do not. We have several concerns with the Company's forecast and methodologies. Our review indicates that the Company's forecast is understated and needs to be adjusted.

Q. Has the Panel developed its own forecast of Con Edison's electric delivery volume in the Rate Year?

A. Yes. A summary of our electric delivery volume forecast and a comparison to the Company's forecast are provided in Exhibit___(SSFP-2).

Q. Please summarize your recommendation regarding the electric delivery volume forecast for Con Edison.

A. We forecast the electric delivery volume for Con Edison to be 53,867 GWh for the Rate Year. Our forecast for the electric delivery volume is reflected in the Rate Year revenue estimated by the Staff Rates Panel. For the reasons discussed later in our testimony, we recommend that our electric delivery volume forecast be adopted by the Commission in this proceeding.

Q. To what degree does the Panel's electric delivery volume forecast differ from that of the Company?

A. Our electric delivery volume forecast is 548 GWh, or 1.0 percent, above the Company's forecast.

Q. Please describe the Company's gas delivery volume forecast.

A. On page 6 of the GFP's updated testimony, the Company forecasted total service area gas delivery volume to be 165,630 thousand dekatherms, or MDt, for the Rate Year.

Q. Do you agree with the Company's gas forecast?

A. No, we do not. We have several concerns with the Company's gas forecast and methodologies.

Q. Has the Panel developed its own forecast for Con Edison's gas delivery volume?

A. Yes, we have.

Q. Please summarize your recommendation.

A. We recommend an increase to Con Edison's gas volume forecast to 165,823 MDt, which equates to an adjustment of 273 MDt, or 0.16 percent of the Company's updated forecast. This adjustment is reflected in the revenue estimate provided in the testimony of the Staff Rates Panel. For the reasons explained later in our testimony, we recommend that our gas volume sales forecast be

adopted by the Commission in this proceeding.

ELECTRIC FORECAST METHODOLOGY

Q. What methodology did the Company use to develop its electric delivery volume forecast?

A. According to pages 8-9 of the EFP's initial testimony, the Company based its forecasts of delivery volumes for all service classifications (SCs), except SCs 5, 6, and 13, on econometric models. An econometric model consists of two parts. The first part is a regression model that relates the delivery volume to a set of independent variables. These independent variables can include the number of customers, real price of electricity, personal income or employment, weather variables, dummy variables, and mobility variables that serve as a proxy to account for the effects of the COVID-19 pandemic. These independent variables normally cannot explain all variations in delivery volume because not all influencing variables are known, quantified, or available with measurable data. In addition, a time series may possess characteristics of non-stationarity or serial correlation, which could result in large

forecasting errors. Therefore, the second part of the econometric model is introduced to capture the remaining variations left by the included independent variables, as well as to address the characteristics that may produce forecasting errors. This part may consist of one or more autoregressive integrated moving average (ARIMA) terms, depending on data patterns, and is statistically determined. The combined model is referred to as an ARIMAX model, where the "X" represents the independent variables included in the model.

Q. Does the Company use the same methodology to forecast its electric customer counts?

A. Yes. Con Edison developed an ARIMAX model for each of the SCs 1, 2, 8, 9, and 12. Similar to the electric volume models, the Company included the mobility, dummy, and ARMA terms as the independent variables, and employment for the service class 1 model.

Q. How is an ARIMAX model estimated?

A. An ARIMAX model is estimated using actual historical data for both the dependent and independent variables. In its update filing,

the Company updated the historical data, forecasted economic variables, and its estimated post-model adjustments through December 2024, except for the SC 12 customer count model in which the historical data ending January 2025 was used to account for higher-than-normal actual customers observed in the fourth quarter of 2024. The Company then used this updated data to produce its updated electric delivery volume and customer forecasts.

Q. How is an ARIMAX model used to forecast delivery volume?

A. After the ARIMAX model is estimated, the relationships between delivery volume and independent variables, along with the coefficients of the ARIMA terms for each service class, are established. The estimated coefficients of the forecasting equation are applied to the forecasted values of the explanatory variables to arrive at the forecast of the dependent variable.

Q. After estimating its ARIMAX models, did the Company make adjustments to the models' forecasts to produce its final forecast?

A. Yes. The Company made various post-model adjustments to account for impacts that may not have been captured by the ARIMAX model. These adjustments included projected changes in sales resulting from energy efficiency, solar generation, steam-to-electric air conditioning conversions, battery storage, standby service, electric vehicles, building electrification, and projected delivery volumes from the development of Hudson Yards.

Q. What methodology did the Panel use to develop its electric delivery volume forecast?

A. Similar to the Company, we used an econometric ARIMAX modeling approach to develop our electric delivery volume forecast. Like the Company, we used the updated data, through December 2024, as provided in Con Edison's responses to DPS-270, DPS-271, DPS-344, DPS-879, DPS-1047, as well as the Company's workpapers provided in the updated filing to produce our forecast.

Q. Given that the Panel and the Company used a similar ARIMAX modeling-based approach to forecast electric customer counts and sales, please explain why the Panel disagrees with the

Company's electric forecasts.

A. One reason the Company's electric forecast should be modified is to address our concerns with some of the independent variables the Company uses in its econometric forecasting models. Specifically, we have concerns with how the Company developed its weather variables and its mobility variables. Thus, even though we and the Company both used a similar ARIMAX modeling-based approach, the use of different independent variable inputs results in different Staff and Company forecasts.

Q. Please explain the adjustments the Company made to its weather variables.

A. As described on page 14 of the EFP's initial testimony, the Company incorporated the impact of climate change on cooling degree days (CDDs) and heating degree days (HDDs) by adjusting normal CDD and HDD in the forecast period based on the results of the Company's 2019 Climate Change and Vulnerability Study (CCVS), filed in Case 22-E-0222. The Company used the 10-year averages of CDD and HDD to represent normal weather. Then, the Company increased the

forecast of normal CDD and decreased the forecast of normal HDD by applying the annual rates of change in CDD and HDD, respectively, based on the results of the Company's CCVS.

Q. What concerns does the Panel have with the adjustments made by the Company to its weather variables?

A. We are concerned that the implied annual rates of change from the Company's Climate Change and Vulnerability Study are based on weather data from different weather stations than the Company used to develop the weather variables for its forecasting effort. In the response to DPS-1051 the Company notes that Central Park's temperature data is used for the HDD and CDD weather variables – however the Central Park weather station data is excluded in the climate change projections due to absence of hourly-scale data – while LaGuardia Airport was used as a surrogate for Central Park in the climate change projection exercise. The use of LaGuardia Airport as a proxy for the weather of Central Park is an apples-to-oranges comparison since subtle temperature differences, especially

between an airport and a park, can have a material impact on the calculated adjustment. Furthermore, we are concerned the date range of the weather data is not reflective of the most recent weather history. As noted on the page 20 of the 2019 CCVS, the date range of the data is from 1976 to 2005, which does not consider the most recent 20 years of weather history, whereas the date range to calculate the CDD and HDD variables in this case uses the most recent 10-year weather history from 2015 to 2024.

Q. Does the Panel agree with applying a weather trend to a 10-year weather normal for purposes of a short-term sales forecast?

A. No, we do not. Based on the concerns stated previously, we do not agree with using a weather trend developed from different weather stations that cannot be verified. For rate cases, the forecasting horizon is only one to three years, as noted on page 14 of the initial testimony of the EFP, the latest 10-year average CDD and HDD are the averages of the most current CDD and HDD and should be representative of the normal weather condition for the forecast without any

need for adjustment. For these reasons, we do not agree with the Company's climate change adjustment to its weather variables.

Q. What does the Panel recommend with respect to weather variables?

A. For its simplicity, we recommend the use of the latest 10-year average as normal weather, as endorsed by the Commission on pages 13-15 of the June 22, 2009, Order Adopting Recommended Decision with Modifications in Cases 08-E-0887, 08-G-0888, and 09-M-0004 for Central Hudson Gas and Electric Corporation (2009 Central Hudson Rate Order). In our volume forecast, we used the non-adjusted CDD and HDD normal weather for 2025-2028, as provided in the Company's response to DPS-271 to develop our electric volume forecast.

Q. Please explain the Panel's concern with the Company's mobility variables.

A. As described on pages 17 to 19 of the EFP's initial testimony, the Company's mobility variables are ratio variables that serve as proxies for the impact of the COVID-19 pandemic on the Company's sales. These variables are

based on Google mobility data that indicate the proportional deviation of daily customer mobility in segments of the economy relative to the median value from the five-week period of January 3 through February 6, 2020, the time immediately before the pandemic began. Mobility in the residential segment impacts SCs 1, 8, and 12; mobility in the retail and recreation segment impacts SC 2; and mobility in the workplace segment impacts SC 9. Thus, the Company created three separate mobility variables based on these three Google-defined segments of the economy.

Q. What concerns does the Panel have with the Company's mobility variables?

A. First, we are concerned that the data comes from a private source. It is not clear that these data have similar quality controls in place as the government-supplied weather and economic data that is typically used for independent variables. Second, the mobility data is not reflective of the mobility of all people in the Company's service territory. The mobility data is only collected from Google Account users who

have opted-in to sharing their location history, which is off by default, thus creating a selection bias issue. Third, the reports use differential privacy which deliberately adds artificial noise to datasets. While this protects privacy, it also introduces a level of imprecision, such as gaps in the identification of mobility, when privacy and data quality thresholds are not met. Fourth, the Google mobility reports were first introduced to account for unprecedented uncertainty caused by restrictions on social and public gatherings due to State executive orders and New York City health mandates – which were largely lifted by June 2021. Therefore, continuing the use of Google mobility reports as a proxy for the impact of COVID-19 adds the potential for misinterpreting normal patterns like public holidays as COVID-19 responses. Furthermore, the baseline period of January 3 to February 6, 2020, is relatively short and might not have fully picked up the seasonal variations or longer-term trends for baseline comparison. Finally, it is not clear why the mobility of a

subset of people in Con Edison's service territory would provide the best indication of the impact of the pandemic on its customers' willingness to consume electric energy.

Q. Do you have specific concerns with the mobility index?

A. Yes, the panel recognizes the Google mobility index may have provided some signal value – during the early stages of the COVID-19 pandemic – when strict health mandates and restrictions on public gatherings created uncertainty in energy usage patterns and customer attrition. Given the following points, the Panel believes the Google mobility index is an inferior variable at best. First, the customer attrition in 2020 related to the pandemic for SCs 1 and 9 recovered by 2023. The average use-per-customer (UPC) returned to pre-pandemic levels by 2023 for SC 1 residential customers, and sales recovered sharply for SC 2 by 2022. SC 9 accounted for the largest drop in sales due to the pandemic impacts, half of which recovered by 2022 where it settled to a "new normal". Second, based on the responses to DPS-271 and

DPS-344 the economy in the Company's service territory largely recovered private non-manufacturing employment (PNEMP) and Real Gross Domestic Product (RGDP) levels by the fourth quarter of 2022, which made the mobility index increasingly irrelevant. Third, any "new normal" in customer behavior patterns is more affectively modelled through economic variables and price sensitivity metrics, which directly reflect consumption decisions. Finally, any residual pandemic impacts not captured by economic fundamentals can be modelled through the ARMA terms in the ARIMA forecasting models, which account for temporary shocks without requiring the use of imprecise mobility metrics, especially given the Rate Year forecast horizon to be near-term.

Q. How does the Panel recommend the Commission address your concerns regarding the Company's mobility variables?

A. The pandemic lockdowns led to a greater decrease in employment in the retail, leisure, and hospitality sectors and thus the degree or impact of lockdown was directly correlated with

these sectors. We included the employment variables from these sectors provided by the Company in response to DPS-271 in the SC 2 and 9 commercial models. We also replaced the mobility index the Company uses in the customer models with the traditional economic variables, as shown in Exhibit___(SSFP-4). This employment data is published by the United States Bureau of Labor Statistics which, as a government source, is more reliable and transparent than private sources. Also, these sector employment forecasts are publicly available and thus do not require the use of the Company's subjective forecasting assumptions.

ELECTRIC POST-MODEL ADJUSTMENTS

Q. Please discuss your post-model adjustments to your electric volume forecast.

A. Energy efficiency (EE), photovoltaic (PV) solar generation, and electric vehicles (EVs), are referred to as distributed energy resources (DERs), and have impacted the actual historic volume data we used to develop our electric volume forecast. Therefore, the impact of historical actual DER savings has already been

reflected in the historical data used to estimate the econometric model used to forecast delivery volume. However, the incremental impact of DERs may not be fully captured in the forecast to the extent that future rate of DER activity is expected to grow beyond the historical rate. For our service class-specific forecasting modelling, this potential discrepancy is analyzed by comparing the historical trend and projected future DERs. In the early years of DERs, when there was not enough historical data to develop a trend to reflect future DERs in a forecast, a post-model adjustment for projected incremental DERs was required. Now that DERs have existed for more than 10 years, there is a sufficient data history to develop a trend in DER activity. In this situation, the projected DERs should be compared with the historical trend to determine whether a post-model adjustment is needed. If the projected DERs follow the same trajectory as the historical trend, no post-model adjustment for DERs is needed. However, if the projected future DERs exceed the historical trend, the

projected DERs incremental to the trend will not be captured in the forecast and should be accounted for through an adjustment to the forecast. To avoid any double count of the impact of DERs, only the projected DERs incremental to the trend should be included in the adjustment. Thus, the Panel's post-model electric adjustments only capture the percentage of the Company's post-model DER adjustments that are incremental to the regression analyses we conducted to determine the historical trends in EE, PV, and EVs. The data used to produce these regression analyses was provided in various attachments to the Company's responses to DPS-598, DPS-879 and DPS-1047 and the results of these regression analyses can be found in Exhibit___(SSFP-6).

GAS FORECAST METHODOLOGY

Q. Please describe the Company's gas sales volume forecasting methodology.

A. As explained on pages 7 to 13 of the GFP's initial testimony, the Company's non-econometric method begins with the Historic Test Year (HTY) actual volume. The Company then makes a number

of adjustments to this base year level of sales in order to produce its rate year sales forecast. Specifically, the Company uses the weather normalized monthly volume data by service class for the HTY, or the 12 months ending September 30, 2024, as the forecasting base, after various adjustments for theft of service, air conditioning volumes, and billing errors. Using the adjusted HTY volume as a base, the Company added the estimated impact of inter-class transfers, customer growth, new customers from oil-to-gas (OTG) conversions and distributed generation, lost customers to electrification, energy efficiency programs, and organic energy efficiency gains. In its updated forecast, as explained on page 2 of the GFP's updated testimony, the Company used calendar year 2024 as the base year but the methodology remained the same.

Q. How did the Company forecast its customer counts?

A. The Company developed its forecast for the five large SC customer counts using an econometric methodology based on historical data, as

explained on pages 16 to 17 of the GFP's initial testimony.

Q. Please briefly describe the Company's weather normalization process.

A. As explained on page 9 of the GFP's initial testimony, the Company applies a regression analysis on nine of the 12 non-summer months of data points of the HTY to establish the relationship between the average actual UPC and the HDD per day for each space heating customer class. This relationship is then applied to the variance between actual HDD and normal HDD and multiplied by total bills to obtain the weather normal adjustment for each month. The water temperature normalization for non-space heating classes was calculated in a similar fashion, including all data for the 12 months.

Q. How does the Company define normal weather?

A. Page 9 of the GFP's initial testimony explains that the Company's normal weather assumption is based on the average weather conditions over 30 years. The Company used this assumption for both normal HDD and normal water temperatures. The Company also applied a climate change

adjustment to its normal weather as described previously in the electric forecast section.

Q. Do you agree with the Company's weather normalization methodology for gas volume forecasting??

A. No, we do not. As an initial matter, using one year's worth of weather and usage data does not give an accurate estimate of weather impacts that deviate from normal. Nine data points associated with the heating season months of the base year is not adequate for a reasonable regression analysis.

Q. What does the Panel recommend for weather normalization?

A. We recommend using a 10-year average to normalize gas delivery volumes, consistent with what we used for the electric volume forecast. On page 14 of the 2009 Central Hudson Rate Order, the Commission determined that a 10-year average weather normalization more reasonably reflects anticipated weather trends than a 30-year average. We also use the most recent 10-year average without applying a climate trend adjustment for the reasons we discussed

previously in the electric forecast section.

Q. Did the Company make other adjustments to HTY gas volumes in preparing its gas volume forecasts?

A. Yes. The Company made other adjustments to the base year volumes to create its volume forecast. These included adjustments for anticipated changes in firm gas sales volumes related to factors such as, but not limited to, net firm gas customer growth, electrification of heating load, and energy efficiency.

Q. What are your concerns with the Company's HTY volumes adjustment process?

A. Since a forecast that relies on adjusting a single base year of data must include adjustments that capture all of the changes in energy usage over time that will carry over to the forecasting period, it may be difficult to accurately reflect all of the impactful changes to sales growth if the base year is either atypical, or if sales growth is not expected to be steady.

Q. Please provide an example that illustrates this concern.

A. Since the pandemic, there have been large changes in economic activity. Our first concern is that the Company's method does not explicitly adjust for the impact of recent swings in the economy on energy usage. Our second concern is that even if it did include such an adjustment, assumptions would have to be made regarding to what extent the base year reflected a typical economic year, and how the forecast should be changed to reflect any anticipated changes in the economy from the HTY to the rate year. In short, the Company's volume adjustment methodology does not capture the dynamics in per-customer usage affected by various factors over time, including economic changes.

Q. Does the Panel have any other concerns with the Company's base year adjustment methodology?

A. Yes. Con Edison argues that the base year was atypical and, therefore, switched from using test year billed sales as the base for making its forecast of expected billed sales to using HTY send out. The Company did this because of the reconciliation of send out and billed sales, as mentioned on page 8 of the GFP's testimony

and in the Company's response to DPS-600.

Q. Was the actual HTY billed gas sales atypical?

A. Yes. The Company states on page 8 of the GFP initial testimony that 2024 was an atypical year for gas sales. It is our understanding that a reconciliation investigation is in progress that attempts to determine why actual billed sales were higher in 2023 than in 2024, even though 2023 was warmer than 2024 and lost and unaccounted for (LAUF) gas was lower in 2024. Given that the reconciliation is still ongoing, the Company decided to use adjusted send out data instead of actual billed sales data as the 2024 base for its forecasting method. When performing this similar forecasting methodology in prior rate cases, the Company has used actual billed sales for the base year's sales.

Q. Does the Panel agree with the Company's decision to substitute 2024 send out sales for 2024 actual billed sales as the base for making its forecast of billed sales?

A. While we do not agree with the Company's method of adjusting only base year sales to make its forecast, we do recognize that 2024 actual

billed gas sales were unreasonably atypical as stated in the Company's November 14, 2024, letter, filed in Case 22-G-0464. As we will discuss later in this testimony, this also requires us to adjust the 2024 data that we use in our alternative forecasting methodology.

Q. Please discuss the Panel's gas forecast and methodology.

A. We use an econometric modelling methodology to develop our forecast for both customer counts and delivery volumes. Similar to our electric forecasting methodology, our forecasting models for gas relate gas delivery volume, or number of customers by service class, to a set of explanatory or independent variables. These independent variables may include economic and demographic variables, HDD, billing days, and other seasonal variables. Our gas volume forecasting models also include the number of customers as an independent variable. All models are estimated using a minimum of ten years of historical data. The estimated models generate forecasts for customer counts and delivery volumes when projected values for the

explanatory variables, such as weather degree days, prices, and other economic variables, are input into the model.

Q. What are the advantages of forecasting gas volumes using an econometric model?

A. Econometric models are estimated in a manner that minimizes errors. The manner in which econometric models include multiple explanatory variables in a single, comprehensive model allows for the robustness of each explanatory variable to be evaluated. Moreover, confidence bands can be calculated around an econometric model's forecast. This type of analysis cannot be performed for the Company's gas volume forecasting methodology, leaving legitimate questions regarding the robustness of the Company's forecasts unanswered.

Q. Please explain the forecasting assumptions for your gas volume models.

A. For weather, we assume the 10-year average of HDDs ending December 31, 2024, as normal weather, without applying the Company's climate change adjustment. These forecasting HDD values were estimated and provided by Con Edison in

response to DPS-271. For economic variables, we used the Moody's forecast provided by Con Edison in response to DPS-271. In addition, because our volume forecast includes total customer counts as an input, we adjusted our customer forecast for the expected customer additions and losses due to OTG conversions before inputting into volume forecasting models. Therefore, our volume forecast already reflects the forecasted impact of OTG conversions and requires no post-model adjustments for these two factors.

GAS POST MODEL ADJUSTMENTS

Q. Did you make any post-model adjustments to your gas volume model forecast?

A. Yes. We made adjustments for the expected changes to our forecast that our models may not have captured. These adjustments reflect the impacts of expected energy efficiency and electrification. Our model includes relevant independent variables and was developed based on more than ten years of historical data, thus requiring no out-of-model adjustments for weather, billing days, customer growth, or

organic energy efficiency. We did, however, make an adjustment to reflect the additional impact of the projected energy efficiency programs in the forecasting period on our gas volume forecast.

Q. How did you make the adjustment to reflect the impact of energy efficiency programs on your gas volume forecast?

A. We first examined the historical trend of energy efficiency. We then adjusted our gas volume forecast for the amount of energy efficiency in excess of the approximated trend, similar to the electric post-model adjustment methodology we discussed previously. The historical energy efficiency data for our trend analysis was provided by the Company in its responses to DPS-865. Our trend analysis for gas energy efficiency is provided in Exhibit___(SSFP-7).

Q. What is the Panel's forecast and the output of your forecasting models?

A. In Exhibit___(SSFP-3), we present a summary of our gas customer and volume forecasts for the Rate Year ending December 2026. Along with our forecast, we also provide a comparison with the

Company's initial forecast and updates. The output of our forecasting models is provided in Exhibit___(SSFP-5).

AGGREGATE ENERGY VOLUME FORECASTS

Q. Does the Company prepare forecasts of total GWh and DTh energy requirements for the aggregate of all its customer classes?

A. Yes.

Q. Please describe the Company's aggregate electric sales volume forecasting methodology.

A. As described on page 35 of the EFP's initial testimony, the Company estimates an econometric model of electric send out.

Q. Why does the Company prepare an electric send out forecast?

A. The Company prepares an electric send out forecast for use in forecasting monthly fuel costs by Company Witness Kimball.

Q. Does the Company have a similar econometric forecast of gas send out?

A. Not that we are aware of. But the Company's Gas Supply Panel does evaluate the Summer Season Supply relative to the Summer Season Demand Balance for the combination of Con Edison and

Orange and Rockland Utilities, Inc. (ORU), as well as its Winter Season Supply relative to the Winter Season Demand Balance for purposes of comparing those companies' firm transportation and supply capability to serve gas demand for firm sales customers' summer and winter seasons under normal weather.

Q. Does the Panel know of other reasons beside forecasting fuel costs and evaluating supply capabilities that warrant top-down aggregate forecasting for both electric and gas delivery volumes?

A. Yes. For example, an aggregate forecast such as the send out forecast, provides a top-down view of the electricity and gas the Company demands. Such forecasts, which are more aggregate in nature than service class or customer group-specific forecasts, can provide a better method to model structural economic impacts on the Company's overall sales and revenues. Thus, aggregate send out forecasts, can also better answer questions about how changes in economic conditions or expected price levels will affect the sales volume. In fact, page 41 of the

Company's June 30, 2016, Distributed System Implementation Plan (DSIP) filed in Case 14-M-0101, mentions the use of such "top-down" forecasts to model market impacts.

Q. Is there another benefit associated with top-down aggregate forecasting?

A. Yes. These aggregate models can also provide checks on the accuracy of the more granular service class-specific models and the reasonableness of the Company's forecasting methodology.

Q. How have top-down models been used by the Commission in the past?

A. Aggregate demand models were used in New York Telephone Company rate cases, such as Case 90-C-0191, to measure the overall impact of price increases on the New York Telephone Company's revenues. The synchronization of top-down and bottom-up forecasts as a means to improve forecast accuracy was also discussed on page 29 of the 2016 Order Adopting Distributed System Implementation Plan Guidance, issued April 20, 2016, in Case 14-M-0101 (2016 DSIP Order).

Q. What is the Company's electric send out volume

forecast.

A. On page 7 of the EFP's initial testimony, the Company forecasted a send out of 58,120 GWh for the Rate Year, which is the 12 months ending December 2026. As shown on page 4 of the EFP's updated testimony, the Company now forecasts a send out of 56,760 GWh for the Rate Year. This is a decrease of 1,360 GWh.

Q. How does the Company's electric send out forecast compare to the summation of its service class forecasts?

A. Exhibit___(EFP-6) shows the Company's forecasted send out for the HTY, Rate Year 1 (RY1), Rate Year 2 (RY2), and Rate Year 3 (RY3). Exhibit___(EFP-7), pages 2 through 5, shows the summation of the Company's electric service class forecasts. The Company's forecasted electric send out for RY1 is 58,120 GWh and the summation of the Company's electric service class forecasts for RY1 is 53,229 GWh. After accounting for anticipated line loss this is a 2.6 percent difference. The difference for RY2 is 3.3 percent and for RY3 is 4.2 percent. In the updated testimony, the percentage

differences for RY1, RY2, and RY3 are 0.1, 0.2, and 0.0 percent, respectively. These percentage differences between the top-down electric send out model and the summation of the electric service class models, in the initial filing, could be indicative of a forecasting error that should be addressed. The Company dramatically improved its forecast between the initial filing and the update.

Q. How does the Company's gas send out forecast compare to the summation of its gas service class forecasts?

A. Since there was no Con Edison-specific send out forecast included in the Company's Gas Supply Panel testimony, only a combined send out forecast for both Con Edison and ORU, and since the Company did not include an econometric send out forecast in its GFP testimony, the Panel was unable to readily compare the Company's send out forecast to its bottom up service class forecasts. As shown in Exhibit__(SSFP-8), Schedule 1, we added expected line losses and interruptible gas sales to the sum of the Company's volumetric service class sales

forecasts. This resulted in 183,000 MDt in RY1, 179,648 MDt in RY2, and 179,178 MDt in RY3 in the initial filing and 182,874 MDt in RY1, 178,865 MDt in RY2, and 177,575 MDt in RY3 in the updated filing.

Q. Did the Company synchronize its send out volume forecast with its "bottom-up" service class forecasts?

A. Not that we are aware of, and not in the manner we described previously.

Q. Has the Panel developed its own forecast for Con Edison's electric and gas send out volumes?

A. Yes. We used econometric modelling to create the Panel's send out forecasts for and electric and gas.

Q. Which forecast should the Commission rely upon since the Panel's send out forecast differs from the sum of its disaggregate service class specific-forecasts?

A. Our intention is to illustrate how aggregate top-down forecasts could be estimated for purposes of economic scenario analysis and the synchronization of top-down and bottom-up models to improve accuracy. We recommend that the

Commission use the Panel's disaggregate sales forecasts for revenue requirement-setting purposes. However, we also recommend that in future rate cases the Company follow the 2016 DSIP Order's directive to synchronize top-down aggregate models and forecasts with its bottom-up service class-specific models and forecasts.

Q. Describe the econometric model the Company used to develop its electric send out forecast.

A. According to page 6 of Exhibit___(EFP-1), the Company included weather variables like HDD and CDD, economic variables such as the real price of electricity, employment in the service territory, a service class-weighted customer index, a COVID-19 mobility variable, and a dummy variable for 2012 quarter 4. In addition to these independent variables, moving average terms and a first difference of all other variables were included.

Q. Please describe the Company's change in methodology in its updated filing.

A. As part of its Corrections and Updates filing, the Company provided a substantively changed model, shown on page 6 of Exhibit___(EFP-9),

which removed the real price of electricity as well as the service class-weighted customer index as explanatory variables. The Company also removed the COVID-19 mobility variable. In addition to changing the model specification, the Company shifted the time period of the historic data upon which the model was estimated to start at quarter 2 of 2000 as opposed to the estimation period of the initially filed model, which started in quarter 1 of 2000.

Q. Do you have concerns with the manner in which the Company structurally changed the specification of the send out model from its initial filing?

A. Yes. The changes in the explanatory variables included in the model were dramatic and raise questions regarding why the Company changed the underlying economic theory behind how its send out model should be specified.

Q. Please discuss how the Panel developed its aggregate send out econometric model for electric sales.

A. We included price as an explanatory variable in our electric send out model since price is an

important driver of the demand for utility service. We also included employment or Gross Domestic Product (GDP) as explanatory economic variables to reflect that overall economic conditions are an important factor driving sales growth.

Q. Have you made other improvements to the Company's electric send out model specification for the purposes of creating a top-down model that can be used for economic scenario analysis and bottom-up model synchronization?

A. Yes. We included different AR and MA terms and a price index. The model estimation period was shortened due to data limitation reasons. The Panel also internalized two variables that had previously been evaluated via post-model adjustments. The DER and building electrification (BE) variables are included in the econometric model to account for the impact of PV, battery energy storage system (BESS), EE, BE, conversions from steam chillers, and EVs over time.

Q. What data did you use to estimate your electric send out model?

A. We use data provided by the Company through the discovery process and data collected from outside sources. The outside sources include the New York Independent System Operator (NYISO), prior rate cases, and EValuateNY. The system-wide average monthly Location Based Marginal Price (LBMP) was calculated from NYISO data. Prior rate cases were used to collect the delivery rate changes for Con Edison. EValuateNY provides historic numbers of EVs on the road in the Company's service territory. CDD, HDD, non-manufacturing employment, historic GWh send out, historic and projected PV GWh generation, EV fleet projections, historic and projected EE, historic and projected BESS, projected steam chiller conversions, and historic and projected BE were all provided by the Company in its responses to DPS-124, DPS-380, DPS-599, and DPS-879.

Q. How is the price index variable for the electric send out forecast used by the Panel different from the Company's real price of electricity variable?

A. The index used in our model is constructed using

an index of the LBMP, the proportion of a customer's 2024 electricity bill attributed to the supply of electricity, a delivery rate price index based on previous rate case determinations, and the Consumer Price Index (CPI).

Q. How does the Panel's electric send out forecast compare to its summation of the service class-level forecasts?

A. In order to do an apples-to-apples comparison, we compared the summation of the service class-level forecasts after accounting for line loss to the send out forecast. Our aggregate electric forecasting model estimates and forecast are shown in Exhibit___(SSFP-8), Schedule 2.

Q. What data did you use to estimate your gas send out model?

A. The Panel used historical send out MDt data from January 2010 to December 2024 that were provided by the Company in response to DPS-315 as the dependent variable in our gas send out model. We used calendar heating degree days, GDP, the log of a price variable, and MMBtu

electrification savings as independent explanatory variables. The Company provided data for heating degree days and GDP and the Panel developed the price variable and electrification savings variables.

Q. How was the price variable in the Panel's gas model created?

A. The price variable was created by using historical real gas prices that were used in the Company's gas volume forecasts. To determine the price, a weighted average was calculated using the Company's sales as the weights. The weights reflect the percent of total sales that each service class represents in a month.

Q. What GDP variable did you use in your aggregate gas volume forecast?

A. We used New York City GDP data provided by the Company in response to DPS-271.

Q. How does the forecast from the Panel's aggregate gas send out model compare to the sum of the your individual service class-specific forecasts?

A. After adding line losses and interruptible gas sales to the Panel's service class-specific

forecasts, a comparison to the aggregate top-down forecast is shown in Exhibit___(SSFP-8), Schedule 3.

Q. Did the Panel address the issue of stationarity as it relates to your aggregate electric and gas models?

A. Yes. The residuals for both the electric send out model and the gas send out model were tested for stationarity. From these tests the models were determined to be stable.

Q. Is there another manner in which you attempted to synchronize your electric send out model to your gas send out model?

A. Yes. A typical post-model adjustment made to the gas and electric service class-specific models relates to BE. This adjustment is intended to measure the increase in electric usage by customers and corresponding decrease in gas usage by those same customers due to switching from gas heating, cooking, drying, and water heating to the electric version of the appliances and equipment that perform these functions. By incorporating a BE variable into both models, and then estimating them via a

system of equations, it is possible to obtain more robust estimates of the impact of BE on both gas and electric customer usage.

Q. Is it possible to estimate these models in a system of equations to more efficiently predict the increases in electric load associated with BE and the corresponding decreases in gas load?

A. Yes. For example, this could be done via a Seemingly Unrelated Regression (SUR) estimation procedure.

Q. Did you perform a SUR estimation of your top down electric and gas models as a system of equations?

A. No. We could not develop a consistent BE variable for the overlapping portion of the Company's gas and electric service territory based on historic data and Company projections. We recommend that the Company perform a system of equations estimation, such as the SUR procedure, for its next rate filing.

Q. How can the accuracy of the aggregate forecasts be evaluated?

A. This is typically done via ex-post simulations and ex-ante comparisons.

Q. Please describe your ex-post forecasting analysis.

A. To perform an ex-post forecast, we use the estimated model to forecast two years of the historical data period upon which the model was estimated. We decided on a two-year test period because we were concerned about the forecast for the HTY and RY1. We then performed standard statistical tests to assess the accuracy of the forecast over the last two years. The results of our ex-post forecasting tests on our electric and gas send-out models are shown in Exhibit___(SSFP-8), Schedule 4.

Q. Please describe your ex-ante forecast analysis.

A. Our ex-ante forecast analysis uses the historic actuals and the forecasted values from prior rate cases. This comparison shows how the Company's prior forecasts relate to the actuals once the forecasting period has passed. The results of the Panel's ex-ante analyses are shown in Exhibit___(SSFP-8), Schedule 4.

Q What is your conclusion from the Panel's ex-post and ex-ante accuracy analyses?

A. The Panel's electric and gas send out models

appear to do well on the ex-post forecast testing. This would suggest that no changes in the specification of those models would be warranted. However, the ex-ante forecasting analysis suggests that both the previous rate case forecasts of Staff and the Company have often been inaccurate, suggesting ongoing structural changes in the demand for electricity and gas which must be continually addressed.

Q. How could the Panel's aggregate send out models be used for economic scenario analysis?

A. For example, Staff's gas send out model indicates a price elasticity of -0.212. Thus, if there was an interest in estimating what the impact on electric sales would be if a dramatic change in the real price of gas were to occur, the aggregate model's price coefficient estimate could be used to quickly determine the impact that supposed price increases would have on gas sales.

Q. Please describe such scenario analyses in more detail.

A. For example, the last year of the LBMPs used to create the electric price index could be held

constant. However, the price index could be increased by 13.4 percent in the electric scenario or 19.1 percent in the gas scenario to reflect the Company's proposed rate increase in its initial filing. Another example could be to decrease the forecast of the GDP variable used in the gas send out model to reflect a recession scenario.

Q. What are the results of your scenario analyses?

A. Our illustrative scenario analysis which uses the Panel's Estimated Aggregate Models and Different Assumptions About Price, GDP and Employment Growth are presented in Exhibit___(SSFP-8), Schedule 5. While these scenario analyses are meant to be illustrative at this point, we recommend that the Company perform more definitive scenario analyses in its next rate case filing.

FORECASTING RELATED FTEs and O&M EXPENSES

Q. Has the Company requested additional employees to perform forecasting-related tasks?

A. Yes. On pages 48 to 51 of the EFP's testimony, the Company describes the need for five

additional full-time equivalents (FTEs) to perform electric forecasting activities. On page 28 of the GFP testimony the Company describes the need for one additional employee for Gas Peak Demand and Volume Forecasting.

Q. Has the Company described the specific duties and start date for these positions?

A. In response to DPS-511 and DPS-512 the Company provided a general description of the duties of these positions, and why the hiring of the FTEs should not be delayed beyond their specified start dates. The Company also provided an organization chart showing where each of these positions fit within the Commodity Forecasting Department's Org Chart.

Q. Do you agree with the need for these six forecasting related FTEs?

A. No. In two instances the duties of the requested Senior Planning Analyst FTE positions overlap with those of other requested forecasting FTEs and the Company has not sufficiently explained why those FTEs' duties could not be performed by one employee. Another Senior Planning Analyst FTE has a duties

description that is unnecessary since NY Cap and Invest is delayed.

Q. Are all three forecasting FTEs you recommend disallowing forecasted to start their employment during the rate year?

A. No. The Company indicated that two of these FTEs would begin in RY1 and one of the FTEs would begin in RY2.

Q. Do the Company's EFP and GFP also request recovery of operations and maintenance (O&M) expenses?

A. Yes.

Q. Are the requests for the recovery of these forecasting O&M expenses reasonable?

A. The Company's responses to DPS-682 and DPS-708 indicate that that the work associated with vendor labor for maintaining the energy management analytics and advanced metering infrastructure (AMI) business analytics overlaps with the work that the forecasting-related FTEs that we recommend disallowing. Allowing this maintenance work via contracted labor would be advantageous in that the vendor labor expenses end when the contracts end. The alternative of

performing the maintenance work via Con Edison FTEs would be disadvantageous to ratepayers as those labor costs would end up being recovered more permanently in base rates.

Q. Does this conclude your testimony at this time?

A. Yes.